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Section: B4=B3-49

Problem Statement:

Vhagar is an AI-driven financial intelligence platform engineered to mitigate cognitive bias and identify market mispricing. By integrating real-time sentiment analysis with volatility divergence, the system identifies irrational market consensus. Its cross-asset surveillance maps ripple effects across supply chains and competitors, while an integrated "Devil's Advocate" chatbot challenges investment theses to minimize behavioral errors. Supplementing these insights, the platform utilizes multi-modal Python-based charting to deliver interactive, visual intelligence, providing investors with a comprehensive toolkit for high-conviction decision-making.

1. The Mispricing Core (Sentiment vs. Volatility Divergence)

What it does: Pulls real-time news sentiment (NLP scoring) and compares it against the stock's Options Implied Volatility (market fear).

The Output: A "Divergence Score" (0–100). If sentiment is euphoric but volatility is low, it flags an irrational market consensus—warning users that the stock is mispriced, rather than just guessing tomorrow's price.

2. The Ripple Effect Mapper (Cross-Asset Surveillance)

What it does: Automatically maps a company's top suppliers and direct competitors. When you search for a stock (e.g., Tesla), it simultaneously fetches breaking news for its supply chain partners (e.g., Panasonic, LG) and rivals.

The Output: A "Domino Heatmap" showing how news affecting secondary tickers could ripple back to impact your primary stock, giving you a 3-hour head start on market moves.

3. The Devil's Advocate Chatbot (Cognitive Bias Slayer)

What it does: A dual-mode integrated chatbot. Mode 1: Answers standard queries via RAG (Retrieval-Augmented Generation). Mode 2 (Unique): You input your personal investment thesis, and the bot actively searches for only the news, analyst downgrades, and macro headwinds that contradict your view.

The Output: A structured, professional rebuttal designed to force you to confront your biases before hitting the "buy" button.

4. Multi-Modal Interactive Charting (Visual Intelligence)

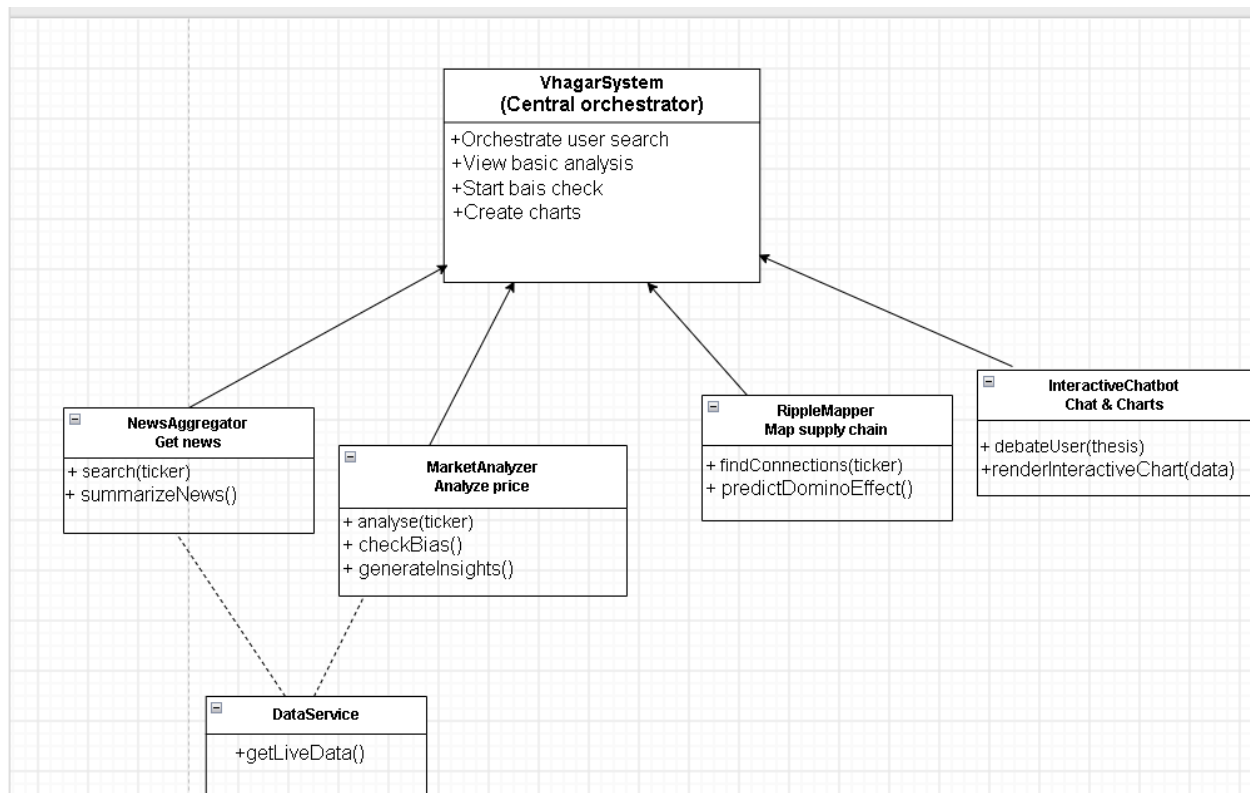
What it does: The chatbot doesn't just respond with text. It dynamically generates Python (Plotly) charts in real-time based on your query.

The Output: If you ask, "Show me the correlation between this news sentiment and the RSI over the last 60 days," the bot renders a fully interactive, hoverable, zoomable chart directly inside the chat interface.

5. The Integrated News & Search Engine

The Foundation: Real-time global news aggregation. When you search a ticker, it pulls the latest headlines, summarizes them, and feeds all the above modules instantly.

CLASS DIAGRAM



Class Descriptions

- **VhagarSystem (Central Orchestrator):** Acts as the main controller for the platform. It coordinates user requests by delegating tasks to specific modules, ensuring a seamless flow between search, analysis, and visualization.
- **NewsAggregator:** Responsible for gathering real-time global news. Its creation enables the system to provide up-to-the-minute data, which serves as the foundation for the platform's analytical modules.

- **MarketAnalyzer:** Performs core financial analysis. It is designed to mitigate bias and identify market mispricing by comparing sentiment against volatility and generating actionable insights.
- **RippleMapper:** Focuses on cross-asset surveillance. It maps supply chains and competitors to predict how news in one sector might create "domino effects" on primary assets.
- **InteractiveChatbot:** Manages the user interface layer. It facilitates the "Devil's Advocate" debates to challenge user biases and renders dynamic, Python-based charts for visual intelligence.
- **DataService:** A foundational utility class that provides the necessary live data streams, ensuring all analytical components have access to accurate, real-time market metrics.